

Lab 2 Experimental Work (Lab EE315)

1. Use the network analyzer
 - a. Measure the S-parameters of the BFU520-based amplifier at the chosen frequency with a $\pm 10\%$ band. Set the supply voltage to the first values between 5V to 12V. Save the touchstone format file in a USB stick.
 - b. Repeat part a with the supply voltage set to another value between 5V to 12V.
 - c. Put the network analyzer in power sweep mode. Set the frequency to your chosen frequency. Sweep the power between -50dBm to -10dBm while observing S21. Set the supply voltage of the amplifier to 5V. Find and record the output 1dB gain compression point using the delta marker.
 - d. Repeat part c with the supply voltage increased in steps of 1V to 12V. Find and record 1dB gain compression point at every step.
2. Use the Spectrum analyzer (refer to the Spectrum Analyzer User Manual)
 - a. Set the supply voltage of the amplifier to the chosen value of preliminary. Set the first generator's frequency to the chosen frequency. Set its amplitude to -40dBm. Set the second generator's frequency 1MHz higher. Set its amplitude to also -40 dBm. Set the center frequency of the spectrum analyzer to the chosen frequency. Set its frequency span to 5 MHz. Measure the magnitudes of third-order products using the marker function of the spectrum analyzer.
 - b. Repeat the measurements when the amplitude of the signal generators set to -30dBm and -20dBm.

In your report, plot the 1dB compression point as a function of the supply voltage. Compare with your expectation from the harmonic balance simulation.

Find the OIP3 for both sidebands while the input levels are varied. Plot OIP3 as a function of input level. Compare these results with those from the harmonic balance simulation.